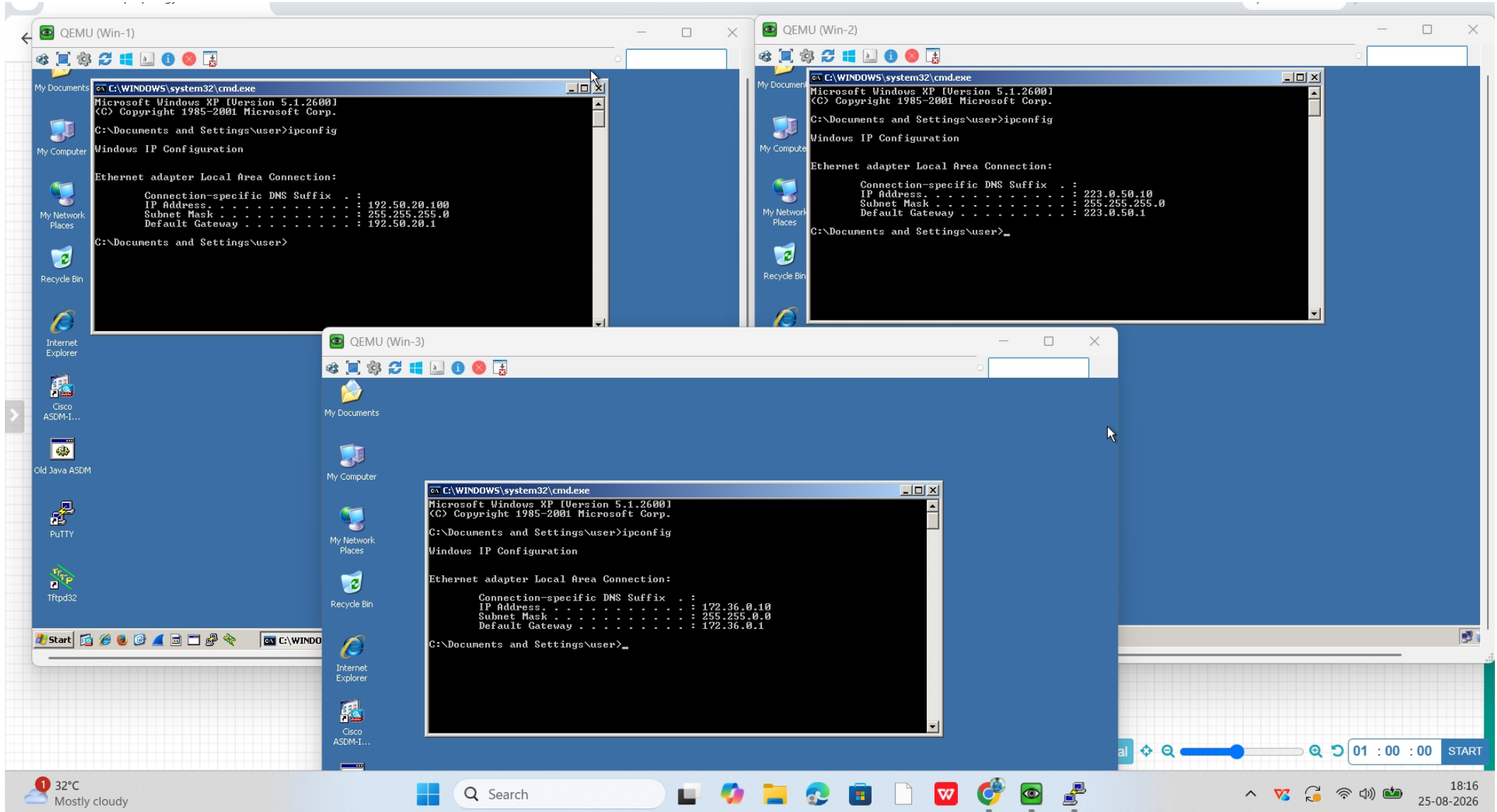


STATIC ROUTE PROJECT -2

ASSIGN THE IP ADDRESSES FOR ALL THE PC'S:-



ASSIGN THE IP ADDRESSES FOR ALL THE ROUTERS INTERFACES:-

PNETLab | Topology

R1

```
R1#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	192.50.20.1	YES	manual	up	up
Ethernet0/1	12.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R1#

R2

```
R2#sh ip int bri
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	12.0.0.2	YES	manual	up	up
Ethernet0/1	26.0.0.1	YES	manual	up	up
Ethernet0/2	23.0.0.1	YES	manual	up	up
Ethernet0/3	24.0.0.1	YES	manual	up	up

R2#

R3

```
R3#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	23.0.0.2	YES	manual	up	up
Ethernet0/1	32.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R3#

R4

```
R4#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Protoc
Ethernet0/0	24.0.0.2	YES	manual	up	up
Ethernet0/1	45.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R4#

33°C Mostly cloudy

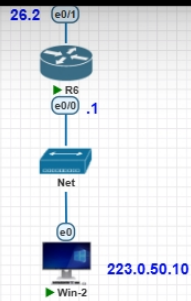
Search

17:58 25-08-2026

00 START

```
R5#sh ip int br
Interface      IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0     172.36.0.1      YES manual up          up
Ethernet0/1     32.0.0.2        YES manual up          up
Ethernet0/2     45.0.0.2        YES manual up          up
Ethernet0/3     unassigned      YES TFTP  down        down
R5#
```

```
R6#sh ip int br
Interface      IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0     223.0.50.1      YES manual up          up
Ethernet0/1     26.0.0.2        YES manual up          up
Ethernet0/2     unassigned      YES TFTP  down        down
Ethernet0/3     unassigned      YES TFTP  down        down
R6#
```



ADD A STATIC ROUTES FOR MISSING ROUTES IN ALL THE ROUTERS:-

The image displays four terminal windows, each representing a different router (R1, R2, R3, R4) in a network topology. The windows show the configuration of static routes and the status of the routing table.

Router R1:

- Legend: o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP, a - application route, + - replicated route, % - next hop override, p - overrides from Pfr
- Gateway of last resort is not set
- 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
- C 12.0.0.0/8 is directly connected, Ethernet0/1
- L 12.0.0.1/32 is directly connected, Ethernet0/1
- S 23.0.0.0/8 [1/0] via 12.0.0.2
- S 24.0.0.0/8 [1/0] via 12.0.0.2
- S 26.0.0.0/8 [1/0] via 12.0.0.2
- S 32.0.0.0/8 [1/0] via 12.0.0.2
- S 45.0.0.0/8 [1/0] via 12.0.0.2
- S 172.36.0.0/16 [1/0] via 12.0.0.2
- C 192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks
- C 192.50.20.0/24 is directly connected, Ethernet0/0
- L 192.50.20.1/32 is directly connected, Ethernet0/0
- S 223.0.50.0/24 [1/0] via 12.0.0.2
- R1#
- R1#
- R1#
- R1#
- R1#

Router R2:

- Legend: + - replicated route, % - next hop override, p - overrides from Pfr
- Gateway of last resort is not set
- 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
- C 12.0.0.0/8 is directly connected, Ethernet0/0
- L 12.0.0.2/32 is directly connected, Ethernet0/0
- C 23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
- C 23.0.0.0/8 is directly connected, Ethernet0/2
- L 23.0.0.1/32 is directly connected, Ethernet0/2
- C 24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
- C 24.0.0.0/8 is directly connected, Ethernet0/3
- L 24.0.0.1/32 is directly connected, Ethernet0/3
- C 26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
- C 26.0.0.0/8 is directly connected, Ethernet0/1
- L 26.0.0.1/32 is directly connected, Ethernet0/1
- S 32.0.0.0/8 [1/0] via 23.0.0.2
- S 45.0.0.0/8 [1/0] via 24.0.0.2
- C 172.36.0.0/24 is subnetted, 1 subnets
- S 172.36.0.0 [1/0] via 23.0.0.2
- S 192.50.20.0/24 [1/0] via 12.0.0.1
- S 223.0.50.0/24 [1/0] via 26.0.0.2
- R2#
- R2#

Router R3:

- Legend: a - application route, + - replicated route, % - next hop override, p - overrides from Pfr
- Gateway of last resort is not set
- S 12.0.0.0/8 [1/0] via 23.0.0.1
- C 23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
- C 23.0.0.0/8 is directly connected, Ethernet0/0
- L 23.0.0.2/32 is directly connected, Ethernet0/0
- S 24.0.0.0/8 [1/0] via 23.0.0.1
- S 26.0.0.0/8 [1/0] via 23.0.0.1
- C 32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
- C 32.0.0.0/8 is directly connected, Ethernet0/1
- L 32.0.0.1/32 is directly connected, Ethernet0/1
- S 45.0.0.0/8 [1/0] via 32.0.0.2
- S 172.36.0.0/16 [1/0] via 32.0.0.2
- S 192.50.20.0/24 [1/0] via 23.0.0.1
- S 223.0.50.0/24 [1/0] via 23.0.0.1
- R3#
- R3#
- R3#
- R3#

Router R4:

- Legend: o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP, a - application route, + - replicated route, % - next hop override, p - overrides from Pfr
- Gateway of last resort is not set
- S 12.0.0.0/8 [1/0] via 24.0.0.1
- S 23.0.0.0/8 [1/0] via 24.0.0.1
- C 24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
- C 24.0.0.0/8 is directly connected, Ethernet0/0
- L 24.0.0.2/32 is directly connected, Ethernet0/0
- S 26.0.0.0/8 [1/0] via 24.0.0.1
- S 32.0.0.0/8 [1/0] via 45.0.0.2
- C 45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
- C 45.0.0.0/8 is directly connected, Ethernet0/1
- L 45.0.0.1/32 is directly connected, Ethernet0/1
- S 172.36.0.0/16 [1/0] via 45.0.0.2
- S 192.50.20.0/24 [1/0] via 24.0.0.1
- S 223.0.50.0/24 [1/0] via 24.0.0.1
- R4#
- R4#
- R4#
- R4#

The bottom of the image shows a Windows taskbar with the date 25-08-2026, time 18:08, and a notification for "Temps to rise Tomorrow".

```

R5
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

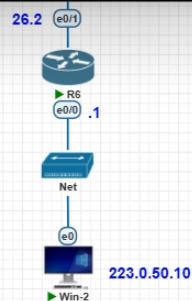
S 12.0.0.0/8 [1/0] via 32.0.0.1
S 23.0.0.0/8 [1/0] via 32.0.0.1
S 24.0.0.0/8 [1/0] via 45.0.0.1
S 26.0.0.0/8 [1/0] via 32.0.0.1
S 32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 32.0.0.0/8 is directly connected, Ethernet0/1
L 32.0.0.2/32 is directly connected, Ethernet0/1
C 45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 45.0.0.0/8 is directly connected, Ethernet0/2
L 45.0.0.2/32 is directly connected, Ethernet0/2
L 172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.36.0.0/16 is directly connected, Ethernet0/0
L 172.36.0.1/32 is directly connected, Ethernet0/0
S 192.50.20.0/24 [1/0] via 32.0.0.1
S 223.0.50.0/24 [1/0] via 32.0.0.1
R5#
R5#
  
```

```

R6
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 26.0.0.1
S 23.0.0.0/8 [1/0] via 26.0.0.1
S 24.0.0.0/8 [1/0] via 26.0.0.1
S 26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 26.0.0.0/8 is directly connected, Ethernet0/1
L 26.0.0.2/32 is directly connected, Ethernet0/1
S 32.0.0.0/8 [1/0] via 26.0.0.1
S 45.0.0.0/8 [1/0] via 26.0.0.1
S 172.36.0.0/16 [1/0] via 26.0.0.1
S 192.50.20.0/24 [1/0] via 26.0.0.1
S 223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C 223.0.50.0/24 is directly connected, Ethernet0/0
L 223.0.50.1/32 is directly connected, Ethernet0/0
R6#
R6#
R6#
  
```



ALL PCS' IP ADDRESSES ARE REACHABLE THROUGH THE ROUTERS:-

The image displays two side-by-side QEMU virtual machines, labeled 'Win-1' and 'Win-2', running a Windows XP environment. The taskbar at the bottom shows a temperature of 32°C, 'Mostly cloudy' weather, and the date '25-08-2026' at '18:20'. A network icon in the taskbar indicates the IP address '223.0.50.10'.

Win-1 (Left VM):

- Command prompt shows the execution of `ipconfig`, displaying the Ethernet adapter 'Local Area Connection' with IP address `192.50.20.100`, subnet mask `255.255.255.0`, and default gateway `192.50.20.1`.
- Subsequent `ping 223.0.50.10` and `ping 172.36.0.10` commands are executed, both showing successful results with 4 packets sent and received, and 0% loss.

Win-2 (Right VM):

- The command prompt shows the execution of `ipconfig`, displaying the Ethernet adapter 'Local Area Connection' with IP address `223.0.50.10`, subnet mask `255.255.255.0`, and default gateway `223.0.50.1`.
- Subsequent `ping 192.50.20.100` and `ping 172.36.0.10` commands are executed, both showing successful results with 4 packets sent and received, and 0% loss.

labs.surennetwork.com/legacy/top

QEMU (Win-3)

Paused

Notifications

!!!!!! Neer

Win-1
192.50.20.100

Net

Win-2
223.0.50.10

Win-3
172.36.0.10

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\user>ipconfig
Windows IP Configuration

Ethernet adapter Local Area Connection:

 Connection-specific DNS Suffix :
 IP Address. : 172.36.0.10
 Subnet Mask : 255.255.0.0
 Default Gateway : 172.36.0.1

C:\Documents and Settings\user>ping 192.50.20.100
Pinging 192.50.20.100 with 32 bytes of data:
Reply from 192.50.20.100: bytes=32 time=3ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124

Ping statistics for 192.50.20.100:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 223.0.50.10
Pinging 223.0.50.10 with 32 bytes of data:
Reply from 223.0.50.10: bytes=32 time=3ms TTL=124
Reply from 223.0.50.10: bytes=32 time=3ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=1ms TTL=124

Ping statistics for 223.0.50.10:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

Start

C:\WINDOWS\system...

1:54 PM

Hot days ahead
32°C

Search

Physical

01 : 00 : 00

START

18:24
25-08-2026